

Computer Networks Study Guide

Chapter 2: Application Layer

2.1 Introduction & Application Architectures

1. What has been the driving force behind the Internet's success according to the chapter?

The creation of numerous useful and entertaining applications that motivate people to make the Internet part of their daily lives

2. List three classic text-based applications from the 1970s and 1980s.

Text e-mail, remote access to computers, file transfers, and newsgroups (any three)

3. What is identified as the "killer application" of the mid-1990s?

The World Wide Web

4. Name two voice/video conferencing applications that emerged in the new millennium.

Skype, Facetime, and Google Hangouts (any two)

5. What type of applications are Facebook, Instagram, and Twitter cited as examples of?

Social networking applications

6. What has enabled the profusion of location-based mobile apps?

The arrival of the smartphone and the ubiquity of 4G/5G wireless Internet access

7. What are the two key aspects of network applications studied in this chapter?

Conceptual and implementation aspects

8. How is an application's architecture different from the network architecture?

The network architecture is fixed and provides a set of services, while the application architecture is designed by the developer and dictates how the application is structured over end systems

9. What are the two predominant architectural paradigms for modern network applications?

The client-server architecture and the peer-to-peer (P2P) architecture

10. In a client-server architecture, what is the role of the server?

It is an always-on host that services requests from many other hosts called clients

11. In the Web application, what is the server and what is the client?

The Web server is the server, and the browser is the client

12. Do clients in a client-server architecture directly communicate with each other?

No, they do not. For example, two browsers do not directly communicate

13. Why can a client always contact a server in a client-server architecture?

Because the server has a fixed, well-known IP address and is always on

14. What is a data center used for in the context of client-server architecture?

To house a large number of hosts to create a powerful virtual server to handle requests from many clients

15. What is the minimal reliance in a P2P architecture?

Minimal (or no) reliance on dedicated servers in data centers

16. In a P2P architecture, what are the peers?

They are intermittently connected hosts (like desktops and laptops controlled by users) that communicate directly with each other

17. What is a compelling feature of P2P architectures?

Self-scalability

18. How does a P2P file-sharing application demonstrate self-scalability?

Although each peer generates workload by requesting files, each peer also adds service capacity to the system by distributing files to other peers

19. What are some challenges faced by P2P applications?

Security, performance, and reliability due to their highly decentralized structure

2.1.2 Processes Communicating

20. What is a process in the context of operating systems?

A process can be thought of as a program that is running within an end system

21. What is the primary interest regarding process communication in this chapter?

How processes running on different hosts (with potentially different operating systems) communicate

22. A network application consists of pairs of _____ that send messages to each other over a network.

Processes

23. In a P2P file-sharing system, what is transferred between processes?

A file is transferred from a process in one peer to a process in another peer

24. How is the client process defined in the context of a communication session?

The process that initiates the communication is labeled as the client

25. How is the server process defined in the context of a communication session?

The process that waits to be contacted to begin the session is the server

26. In P2P file sharing, if Peer A asks Peer B to send a file, which peer is the client?

Peer A

27. What is the software interface through which a process sends and receives messages?

A socket

28. The socket is the interface between which two layers?

The application layer and the transport layer within a host

29. What is another name for the socket?

The Application Programming Interface (API) between the application and the network

30. What two controls does an application developer have on the transport-layer side of the socket?

(1) The choice of transport protocol, and (2) the ability to fix a few transport-layer parameters like maximum buffer and segment sizes

31. What two pieces of information are needed to identify a receiving process?

(1) The IP address of the host, and (2) a port number that specifies the receiving process/socket

32. What is the purpose of a port number?

To identify the specific application process (socket) within the destination host

33. What is the well-known port number for a Web server?

Port 80

34. What is the well-known port number for an SMTP mail server?

Port 25

2.1.3 Transport Services Available to Applications

35. What are the four dimensions along which transport-layer services can be classified?

Reliable data transfer, throughput, timing, and security

36. What is reliable data transfer?

A service that guarantees that data sent by one end of the application is delivered correctly and completely to the other end

37. Give an example of an application that requires reliable data transfer.

Electronic mail, file transfer, remote host access, Web document transfers, or financial applications (any one)

38. What are loss-tolerant applications?

Applications that can tolerate some amount of data loss, such as multimedia applications like conversational audio/video

39. Define throughput in the context of a communication session.

The rate at which the sending process can deliver bits to the receiving process

40. What are bandwidth-sensitive applications?

Applications that have specific throughput requirements

41. What are elastic applications?

Applications that can make use of as much, or as little, throughput as happens to be available (e.g., e-mail, file transfer)

42. Give an example of an application that requires timing guarantees.

Internet telephony, virtual environments, teleconferencing, or multiplayer games

43. How can a transport protocol provide security?

By encrypting all data transmitted by the sending process and decrypting it at the receiving process, providing confidentiality

2.1.4 Transport Services Provided by the Internet

44. What are the two transport protocols available to Internet applications?

UDP (User Datagram Protocol) and TCP (Transmission Control Protocol)

45. What two services does TCP's service model include?

Connection-oriented service and reliable data transfer service

46. What is the purpose of the TCP handshaking procedure?

To exchange transport-layer control information before application-level messages begin to flow, alerting the client and server to prepare for packets

47. After the handshake, what is said to exist between the two processes?

A TCP connection

48. What does TCP's reliable data transfer service guarantee?

That all data sent will be delivered without error and in the proper order

49. What is the purpose of TCP's congestion-control mechanism?

It is a service for the general welfare of the Internet, throttling a sending process when the network is congested

50. Is UDP connection-oriented or connectionless?

Connectionless

51. Does UDP provide reliable data transfer?

No, it provides an unreliable data transfer service

52. Can messages arrive out of order with UDP?

Yes

53. Does UDP have a congestion-control mechanism?

No

54. Which services (reliable data transfer, throughput, timing, security) are not provided by today's Internet transport protocols?

Throughput and timing guarantees

55. Why can time-sensitive applications like Internet telephony still run on the Internet?

Because they have been designed to cope, to the greatest extent possible, with the lack of timing and throughput guarantees

56. Why do Internet telephony applications usually prefer to run over UDP?

To circumvent TCP's congestion control mechanism and packet overheads, as they can tolerate some loss but require a minimal rate

57. Why might an Internet telephony application use TCP as a backup?

Because many firewalls are configured to block (most types of) UDP traffic

2.1.5 Application-Layer Protocols

58. What does an application-layer protocol define?

How an application's processes, running on different end systems, pass messages to each other

59. List the four specific things an application-layer protocol defines.

The types of messages exchanged, the syntax of the messages, the semantics of the fields, and the rules for determining when and how a process sends messages and responds to messages

60. What is the application-layer protocol for the Web?

HTTP (HyperText Transfer Protocol)

61. What is the difference between a network application and an application-layer protocol?

An application-layer protocol is only one piece (albeit important) of a network application. The application includes other components like document standards, clients, and servers

62. What is the application-level protocol used by Netflix?

DASH (Dynamic Adaptive Streaming over HTTP)

2.1.6 Network Applications Covered

63. List the five important applications discussed in detail in this chapter.

The Web, electronic mail, DNS (directory service), P2P applications, and video streaming

64. Why is the Web discussed first?

Because it is enormously popular and its protocol (HTTP) is straightforward and easy to understand

65. Why is e-mail considered more complex than the Web?

Because it makes use of not one but several application-layer protocols

66. How do most users interact with DNS?

Indirectly through other applications (like the Web, file transfer, and e-mail)

2.2 The Web and HTTP

67. What was the first Internet application that caught the general public's eye?

The World Wide Web

68. What is the heart of the Web?

The HyperText Transfer Protocol (HTTP)

69. HTTP is implemented in which two programs?

A client program and a server program

70. What is a Web page (or document) composed of?

Objects

71. What is a Web object?

A file, such as an HTML file, JPEG image, or video clip, that is addressable by a single URL

72. What is the base HTML file?

The main HTML file of a Web page that references the other objects in the page

73. What are the two components of a URL?

The hostname of the server and the object's path name

74. What underlying transport protocol does HTTP use?

TCP

75. What does it mean that HTTP is a "stateless protocol"?

The server sends requested files to clients without storing any state information about the client

76. What are the two types of connections HTTP can use?

Non-persistent connections and persistent connections

77. In non-persistent HTTP, how many objects can be sent over a single TCP connection?

Exactly one request message and one response message (one object)

78. What is RTT?

Round-Trip Time, the time it takes for a small packet to travel from client to server and back to the client

79. For a non-persistent connection, roughly what is the total response time for receiving a base HTML file?

Two RTTs plus the transmission time of the file at the server

80. What are two shortcomings of non-persistent connections?

(1) A new connection must be set up for each object, requiring buffers and variables. (2) Each object suffers a delivery delay of two RTTs

81. How does HTTP with persistent connections work?

The server leaves the TCP connection open after sending a response, and subsequent requests and responses between the same client and server can be sent over the same connection

82. What is pipelining in the context of persistent connections?

The client can make back-to-back requests without waiting for replies to pending requests

83. What is the default mode of HTTP?

Persistent connections with pipelining

2.2.3 HTTP Message Format

84. What are the two types of HTTP messages?

Request messages and response messages

85. What is the first line of an HTTP request message called?

The request line

86. What are the subsequent lines in an HTTP request message called?

Header lines

87. What three fields are in the request line?

The method field, the URL field, and the HTTP version field

88. What is the most common HTTP request method?

GET

89. What is the purpose of the Host: header line?

It specifies the host on which the object resides, which is required by Web proxy caches

90. What does the Connection: close header line tell the server?

That the client doesn't want a persistent connection; the server should close the connection after sending the requested object

91. What is the purpose of the User-agent: header line?

It specifies the browser type, allowing the server to send different versions of the same object to different user agents

92. When is the entity body in an HTTP request message used?

It is used with the POST method, containing what the user entered into form fields

93. How does a form using the GET method include inputted data?

It includes the data in the requested URL itself

94. What are the HEAD, PUT, and DELETE methods used for?

HEAD is similar to GET but asks the server not to return the requested object (used for debugging). PUT is used to upload an object to a server. DELETE is used to delete an object on a server

95. What are the three sections of an HTTP response message?

An initial status line, header lines, and an entity body

96. What three fields are in the status line?

The protocol version field, a status code, and a corresponding status message

97. What does the Last-Modified: header line indicate?

The time and date when the object was created or last modified. It is critical for object caching

98. What is the meaning of the status code and phrase 200 OK?

The request succeeded and the information is returned in the response

99. What is the meaning of the status code and phrase 404 Not Found?

The requested document does not exist on the server

2.2.4 User-Server Interaction: Cookies

100. Why is HTTP's statelessness a problem for some Web sites?

Because they often want to identify users to restrict access or serve content as a function of user identity

101. What technology allows sites to keep track of users?

Cookies

102. What are the four components of cookie technology?

(1) A cookie header line in the HTTP response message; (2) a cookie header line in the HTTP request message; (3) a cookie file kept on the user's end system; (4) a back-end database at the Web site

103. What header line does a server use to set a cookie on a user's browser?

Set-cookie:

104. What header line does a browser use to send a cookie back to a server?

Cookie:

105. What is one potential downside of cookies?

They can be considered an invasion of privacy, as a Web site can learn a lot about a user and potentially sell that information

2.2.5 Web Caching

106. What is a Web cache also known as?

A proxy server

107. How does a Web cache satisfy a request?

It satisfies HTTP requests on the behalf of an origin Web server by storing copies of recently requested objects

108. What are two reasons for deploying Web caches?

(1) To reduce response time for a client request. (2) To reduce traffic on an institution's access link to the Internet

109. How does a Web cache act as both a server and a client?

It is a server to the requesting browser and a client to the origin server

110. What problem does caching introduce?

The copy of an object in the cache may be stale (out of date)

111. What HTTP mechanism allows a cache to verify its objects are up to date?

The conditional GET

112. What two conditions make an HTTP request a conditional GET?

(1) The request uses the GET method, and (2) the request includes an If-Modified-Since: header line

113. If an object has not been modified, what status code does the server return in response to a conditional GET?

304 Not Modified

2.2.6 HTTP/2

114. What was the primary goal of HTTP/2?

To reduce perceived latency

115. How does HTTP/2 achieve its primary goal?

By enabling request and response multiplexing over a single TCP connection

116. Does HTTP/2 change the HTTP methods, status codes, or URLs?

No

117. What is Head-of-Line (HOL) blocking in HTTP/1.1?

When a large object at the head of a TCP connection's queue blocks smaller objects behind it from being delivered

118. How did HTTP/1.1 browsers work around HOL blocking?

By opening multiple parallel TCP connections

119. How does HTTP/2 solve the HOL blocking problem?

By breaking each message into small frames and interleaving (multiplexing) the request and response messages on the same TCP connection

120. What is another key feature of HTTP/2 besides framing?

It uses binary encoding for frames, making them more efficient to parse

121. What is message prioritization in HTTP/2?

It allows a client to assign weights to its requests, so the server can send higher-priority responses first

122. What is server push in HTTP/2?

The ability for a server to send multiple responses for a single client request, pushing additional objects to the client before the client explicitly requests them

123. What transport protocol is HTTP/3 designed to operate over?

QUIC, which is implemented over UDP

2.3 Electronic Mail

124. What are the three major components of the Internet mail system?

User agents, mail servers, and the Simple Mail Transfer Protocol (SMTP)

125. What is the role of a user agent?

It allows users to read, reply to, forward, save, and compose messages (e.g., Outlook, Gmail)

126. Where is a recipient's mailbox located?

On a mail server

127. What is the principal application-layer protocol for Internet electronic mail?

SMTP (Simple Mail Transfer Protocol)

128. When does a mail server act as an SMTP client?

When it is sending mail to other mail servers

129. What is a key restriction of SMTP?

The body of all mail messages must be in simple 7-bit ASCII

130. Outline the basic steps of Alice sending an email to Bob using SMTP.

1. Alice composes message with her UA.
- 2) UA sends message to her mail server.
- 3) Alice's mail server (SMTP client) opens TCP connection to Bob's mail server (SMTP server).
- 4) SMTP client sends message.
- 5) Bob's server places message in his mailbox.
- 6) Bob uses his UA to read the message

131. Does SMTP typically use intermediate mail servers?

No, it usually uses a direct TCP connection between the sender's and receiver's mail servers

132. What separates the header and the body of an email message?

A blank line (CRLF)

133. What are two common ways for a user to retrieve email from a mail server?

Using HTTP (for Web-based email) or using the Internet Mail Access Protocol (IMAP)

134. Why can't a user agent use SMTP to retrieve messages from a mail server?

Because obtaining messages is a pull operation, while SMTP is a push protocol

2.4 DNS

135. What are the two main ways to identify an Internet host?

By a hostname and by an IP address

136. What is the main task of the Domain Name System (DNS)?

To translate hostnames to IP addresses

137. What two things is the DNS?

(1) A distributed database implemented in a hierarchy of DNS servers, and (2) an application-layer protocol

138. What transport protocol and port does DNS typically use?

UDP, port 53

139. Why would a centralized DNS database be a bad design? (Name two reasons)

A single point of failure, inability to handle the traffic volume, distant centralized database leading to delays, and difficult maintenance

140. What are the three classes of DNS servers in the hierarchy?

Root DNS servers, Top-Level Domain (TLD) DNS servers, and Authoritative DNS servers

141. What is the role of root DNS servers?

They provide the IP addresses of the TLD servers

142. What is the role of TLD servers?

They provide the IP addresses for authoritative DNS servers

143. What is the role of authoritative DNS servers?

They hold the DNS records that map the names of an organization's hosts to IP addresses

144. What is a local DNS server?

A DNS server provided by an ISP that acts as a proxy, forwarding queries into the DNS server hierarchy. It is "close to" the host

145. In a typical DNS lookup for `gaia.cs.umass.edu` from `cse.nyu.edu`, how many DNS messages are sent in the example without caching?

Eight messages (four query and four reply)

146. What is the difference between a recursive query and an iterative query?

In a recursive query, the queried server obtains the mapping on the requesting host's behalf. In an iterative query, the server replies directly with the best answer it has (which might be a referral to another server)

147. What is DNS caching and why is it important?

It is when a DNS server stores mappings in its local memory after receiving a reply. It improves delay performance and reduces the number of DNS messages worldwide

2.5 Peer-to-Peer File Distribution

148. What is a key characteristic of a P2P architecture?

Minimal (or no) reliance on always-on infrastructure servers; peers communicate directly

149. In P2P file distribution, how do peers assist the server?

Each peer can redistribute any portion of the file it has received to any other peers

150. What is the most popular P2P file distribution protocol as of 2020?

BitTorrent

151. In the client-server distribution time equation $D_{cs} = \max\{NF/u_s, F/d_{min}\}$, what does the term NF/u_s represent?

The time for the server to send the file to all N peers

152. In the client-server distribution time equation $D_{cs} = \max\{NF/u_s, F/d_{min}\}$, what does the term F/d_{min} represent?

The time for the slowest peer to download the file

153. Why does the distribution time for a client-server architecture increase linearly with N ?

Because the server must send a copy of the entire file to each of the N peers

154. What are the three observations that lead to the P2P distribution time equation $D_{P2P} = \max\{F/u_s, F/d_{min}, NF/(u_s + \sum u_i)\}$?

(1) Server must send each bit at least once: F/u_s . (2) Slowest peer download time: F/d_{min} . (3) Total system upload capacity must deliver NF bits: $NF/(u_s + \sum u_i)$

155. Why are P2P architectures self-scaling?

Because peers are not just consumers but also redistributors of content, adding capacity to the system as more peers join

156. In BitTorrent, what is a collection of peers participating in the distribution of a file called?

A torrent

157. What are the pieces of a file called in BitTorrent?

Chunks

158. What is the role of the tracker in a BitTorrent torrent?

It keeps track of the peers participating in the torrent. When a peer joins, it provides that peer with a list of other peers

159. What is the "rarest first" technique?

A peer requests the chunks that are the rarest (have the fewest copies) among its neighbors first, to equalize the chunk distribution in the torrent

160. How does a BitTorrent peer decide which neighbors to send chunks to?

It prioritizes the four neighbors that are currently supplying data to it at the highest rate (these are "unchoked")

161. What is the purpose of "optimistic unchoking"?

Every 30 seconds, a peer randomly chooses another neighbor and sends it chunks. This allows new peers to get chunks and discover better trading partners

2.6 Video Streaming and CDNs

162. What approximate percentage of Internet traffic did streaming video account for in 2020?

About 80%

163. In HTTP streaming, where is the video file stored?

As an ordinary file on an HTTP server

164. How does the client application in HTTP streaming manage playback?

It collects bytes in a buffer. Once the buffer exceeds a threshold, it begins playback, periodically grabbing frames from the buffer

165. What does DASH stand for?

Dynamic Adaptive Streaming over HTTP

166. How does DASH work?

The video is encoded into multiple versions at different bit rates. The client dynamically requests chunks from different versions based on the current available bandwidth

167. What does the manifest file in DASH provide?

It provides a URL for each version of the video along with its bit rate

168. What are three problems with streaming all videos from a single massive data center?

(1) Long paths may have bottlenecks, causing freezing. (2) Popular videos are sent repeatedly over the same links, wasting bandwidth. (3) It's a single point of failure

169. What does a CDN do?

It manages servers in multiple geographically distributed locations, stores copies of content (like videos) in these servers, and directs user requests to a suitable CDN location

170. What is the difference between a private CDN and a third-party CDN?

A private CDN is owned by the content provider itself (e.g., Google's CDN for YouTube). A third-party CDN distributes content on behalf of multiple content providers

2.7 Socket Programming & 2.8 Summary

171. A developer's main task when creating a network application is to write code for which two parts?

The client program and the server program

172. What do the client and server processes use to communicate with each other?

Sockets

173. The material in this chapter added substance to the definition of a protocol. What are the key elements?

The format and the order of messages exchanged, as well as the actions taken on transmission/receipt

174. Which layer does the next chapter (Chapter 3) examine?

The Transport Layer

Comprehensive & In-Depth Questions

175. Explain the analogy of a process and a socket used in the chapter.

A process is like a house, and its socket is like its door. A process sends a message out its door (socket), and the network infrastructure delivers it to the destination process's door (socket)

176. Why might an application developer choose UDP over TCP despite its lack of reliability?

If the application is loss-tolerant but requires low latency or a specific transmission rate, and wants to avoid TCP's congestion control and connection setup overhead (e.g., Internet telephony)

177. How does the use of a Web cache reduce traffic on an institution's access link?

If a requested object is in the cache, it is returned directly from the cache within the institutional network, without using the access link to the Internet

178. Describe the sequence of DNS servers contacted when resolving a hostname like `www.amazon.com`.

The client's local DNS server contacts a root server, which refers it to a .com TLD server. The TLD server refers it to an authoritative server for amazon.com, which then returns the IP address for www.amazon.com

179. How does BitTorrent's trading algorithm encourage fair and efficient sharing?

By using a tit-for-tat strategy where peers upload to those who are uploading to them at the highest rates, and by using optimistic unchoking to discover new partners and help new peers join the system

180. How does DASH provide a better user experience than simple HTTP streaming?

It allows the client to adapt to changing network conditions by seamlessly switching between different quality versions of the video, minimizing rebuffering events

181. Contrast the client-server and P2P architectures for file distribution in terms of scalability.

Client-server scalability is limited by the server's upload capacity (NF/u_s), causing distribution time to grow linearly with N . P2P scalability is superior because the total system capacity increases as peers join ($u_s + \sum u_i$), keeping distribution time low even for large N

182. What is the significance of the Last-Modified and If-Modified-Since headers in the context of Web caching?

They enable the conditional GET mechanism. The cache stores the Last-Modified date from a previous response. On a subsequent request for the same object, it sends an If-Modified-Since header with that date. If the object hasn't changed, the server replies with 304 Not Modified, saving bandwidth

183. Why was framing introduced in HTTP/2?

To solve the Head-of-Line (HOL) blocking problem present in HTTP/1.1 when using a single TCP connection. By breaking messages into frames and interleaving them, multiple responses can be delivered in parallel, preventing a large or slow response from blocking others

184. What problem does SMTP's 7-bit ASCII restriction cause for modern email, and how is it overcome?

It prevents the direct sending of binary data (like images or attachments). This is overcome by encoding the binary data into ASCII before sending (e.g., using Base64 encoding) and decoding it upon receipt

185. In the context of DNS, what is the role of the local DNS server?

It acts as a proxy for the host. When a host makes a DNS query, it is sent to the local DNS server, which then handles the iterative process of querying the root, TLD, and authoritative servers on the host's behalf. It also caches results for future queries

186. How does the "rarest first" chunk selection policy in BitTorrent benefit the overall health of a torrent?

It ensures that rare chunks are replicated more quickly across the swarm, reducing the chance that the last few chunks needed by any peer become unavailable. This increases the system's robustness and download efficiency for everyone

187. Explain how a CDN improves the performance of video streaming.

By storing copies of videos in servers located geographically closer to users, a CDN reduces the number of links and ISPs between the server and the client. This decreases the likelihood of encountering a bottleneck link, leading to higher throughput, lower latency, and a smoother streaming experience

188. Why is the three-way handshake a significant component of the total response time in non-persistent HTTP?

Because it takes one RTT to complete the first two parts of the handshake before the HTTP request can even be sent. This setup time is incurred for every object in non-persistent connections

189. How does the definition of client and server in a P2P system differ from that in a pure client-server system?

In a pure client-server system, the roles are fixed: the server is always on and services requests, and the client initiates contact. In a P2P system, a single peer's process can act as both a client (when downloading a file from another peer) and a server (when uploading a file to another peer) simultaneously or at different times

190. What is the fundamental reason why P2P file distribution is more scalable than client-server distribution?

The resources (upload bandwidth) of the peers themselves are harnessed to assist in the distribution process. As more peers join the system, the total capacity for distributing the file increases, whereas in client-server, the server's capacity remains fixed

191. Describe the journey of an email message from composition to reading, mentioning the key protocols involved.

1. Composed in sender's User Agent (UA).
- 2) Sent via SMTP (or HTTP) to sender's mail server.
- 3) Sender's mail server (SMTP client) uses SMTP to send the message to the recipient's mail server (SMTP server).
- 4) Message is deposited in the recipient's mailbox on that server.
- 5) Recipient uses their UA to retrieve the message using IMAP or HTTP

192. How does the use of persistent connections in HTTP/1.1 improve performance compared to non-persistent connections?

It eliminates the TCP connection setup and teardown overhead (2 RTTs per object) for all but the first object on a page. It also allows for pipelining, where multiple requests can be sent back-to-back without waiting for individual responses

193. What is the purpose of the Server header in an HTTP response and the User-agent header in an HTTP request?

They identify the software being used. The Server header tells the client what Web server software is running, and the User-agent header tells the server what client software (browser) is making the request. This can be used for compatibility or analytics

194. Why is DNS crucial for the operation of other application-layer protocols like HTTP and SMTP?

Because these protocols ultimately need to communicate with a process on a specific host identified by an IP address. DNS provides the essential service of translating the human-readable hostnames used in URLs and email addresses into the IP addresses required for routing packets

195. In the BitTorrent protocol, what is the purpose of having both "unchoked" and "optimistically unchoked" peers?

"Unchoked" peers (the top uploaders) create a stable, efficient data exchange based on reciprocity (tit-for-tat). "Optimistically unchoked" peers allow for the exploration of new, potentially better trading partners and help new peers with no data to trade get started

196. How does the concept of a "socket" abstract the complexities of network communication for an application developer?

The socket provides a simple interface (like a door) for the application process to send and receive messages. The developer doesn't need to worry about the intricate details of how TCP ensures reliability or how IP routes packets through the network; they just write to and read from the socket

197. What is the key trade-off between using TCP and UDP for a new application?

TCP provides convenient services (reliability, congestion control, flow control) at the cost of potential delays and overhead. UDP provides minimal services, giving the application developer full control over timing and data transfer, but requiring them to build any necessary reliability or congestion control themselves

198. How does the hierarchical structure of the DNS database contribute to its scalability and manageability?

It delegates responsibility. The root servers only need to know about TLD servers. Each TLD server only needs to know about authoritative servers in its domain. Each organization only manages its own authoritative DNS records. This decentralization prevents any single entity from having to manage all Internet hosts

199. Explain how HTTP/2's server push can reduce page load times.

By analyzing the base HTML page, the server can anticipate which objects (like CSS, JavaScript, or images) the client will need to render the page fully. It can then push these objects

to the client before the client requests them, eliminating the RTT and request time for those objects

200. Summarize the key reason why the application layer is a good starting point for studying networking.

Because network applications are the driving force behind the Internet and are the aspects most visible and familiar to users. Understanding applications first provides motivation and context for delving into the underlying protocols in the lower layers

End of Study Guide

Wish you all the best!

Fathy